

Vedant Sanjay Jadhav

AI Engineer | LLMs · Agentic AI · Multimodal AI
vedantjadhav1414@gmail.com +91-7410036328

vedantjadhav-ai.vercel.app
github.com/VedantJadhav701
linkedin.com/in/vedantjadhav-ai ORCID

Education

B.Tech. (AI & ML) Pimpri Chinchwad University, Pune 8.10 CGPA 2023–27

Experience

Tech Mahindra – AI Engineer Intern Jun. 2026 – Present, Pune

- Led the AI team as a Young AI Engineer on the development of **Indus**, contributing to model experimentation, evaluation, AI system design, and technical implementation.
- Worked on **DSLMS**, **VLMS**, and Generative AI systems, building pipelines spanning data prep, retrieval, inference, evaluation, and deployment; worked with **NVIDIA Nemotron 3.5** for LLM experimentation and system development under practical compute constraints.
- Designed and implemented **RAG**, semantic retrieval, and agentic AI workflows using Python, PyTorch, Hugging Face, LangGraph, FastAPI, and Ollama.

DPulseAI Pvt. Ltd. – AI Engineer Intern Oct. 2025 – Apr. 2026, Pune

- Built production Generative AI and RAG systems, improving retrieval accuracy by **35%** and reducing latency by **45%** through hybrid chunking and reranking strategies.
- Architected Dockerized LLM inference pipelines with CI/CD and canary deployments, cutting deployment time to **under 10 minutes** using GitHub Actions and Docker Registry.
- Implemented KV-cache quantization and post-deployment drift monitoring, reducing inference memory requirements and improving LLM serving efficiency on consumer-grade hardware.

Selected AI Projects

SparkAI-50M-A100 – Hybrid Mamba-Transformer LLM Hugging Face

- Designed and trained a **50M-parameter LM from scratch** using a hybrid **Mamba-2 + GQA** architecture on a single NVIDIA A100 80GB GPU, implementing tokenization, training, checkpointing, and evaluation.

Vidya Educational LLM – Multilingual Educational AI Live Demo

- Developed a **1.7B-parameter multilingual Educational LLM** fine-tuned for NCERT Science and Mathematics across **11 Indian languages**, achieving **93.3% aggregate accuracy** on a custom 64-question, 8-script, 4-domain evaluation suite; deployed on Hugging Face Spaces.

WinFix OmniAgent – Agentic Windows Diagnostics GitHub

- Built an **evidence-driven agentic troubleshooting system** with planner-specialist TaskGraph execution, domain-specific diagnostic agents, safety-gated tool execution, and a real-time Gradio control console.

AutoStack AI – Autonomous ML Lifecycle Platform Live Demo

- Built a **production-first AutoML platform** that transforms raw CSV data into optimized models and low-latency REST APIs in **under 60 seconds**, automating profiling, feature engineering, tuning, and deployment.
- Implemented parallel hyperparameter optimization across **Gradient Boosted Trees and Stacking Regressors**, with SHAP-based explainability, sub-10ms inference, and statistical drift monitoring.

TinyVLM – Lightweight Vision Language Model GitHub

- Built a lightweight **VLM** integrating a vision encoder with a compact LM for image captioning and visual question answering, implementing feature projection, cross-modal alignment, and instruction-following generation.

Research & Publications

- [Published, IJEMS 2025 – Scopus, Best Paper Award] *Ensemble and Hybrid ML Approaches for Renewable Energy Forecasting* – DOI
- [Preprint] *PhysicalMamba: State Space Models for Physical Planning and Robotic Task Execution* – DOI
- [Preprint] *RecursiveMAS: A Recursive Multi-Agent Latent Coordination Framework for Embodied Task Optimization* – DOI
- [Under Review] *Small Language Models for Clinical Reasoning and Medical Decision Alignment*

Technical Skills

- Languages:** Python, C/C++
- LLM & Generative AI:** LLMs, SLMs, DSLMs, VLMS, Pretraining, Fine-Tuning (LoRA/QLoRA), RAG, Agentic AI, Multilingual NLP, Inference Optimization
- Model Architectures:** Transformers, Mamba/Mamba-2, GQA, CNNs, RNNs, LSTMs, YOLOv8
- AI Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, Ollama, CrewAI, MCP, Tool Calling
- AI Systems & MLOps:** FastAPI, Docker, MLflow, Git, GitHub Actions/CI-CD, Model Serving, LLMOps, Drift Monitoring, Canary Deployment
- Data & Retrieval:** Qdrant, ChromaDB, FAISS, PostgreSQL, MongoDB, Pandas, NumPy

Achievements

- 1st Place** – Code4Society Hackathon 2026
- Best Research Paper Award** – ICCTVB-25 (Scopus Indexed)
- 3rd Place** – National AI Hackathon, IIT Indore
- 1st Place** – CodeApex 24-hr Hackathon, VIT
- 3rd Place** – National DevCraft Hackathon, IIT Indore